

Chapter 3: Indefinite Integrals

National High School of Mathematics

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1. Indefinite Integrals

1.1 Definition of Primitive

Let f be a real function defined on a closed interval $[a, b]$, and let $F : [a, b] \rightarrow \mathbb{R}$ be a function. We say that F is a **primitive** of f if:

$$F'(x) = f(x).$$

For $f(x) = \frac{1}{x}$ on $1 \leq x \leq 5$, a primitive of f is:

$$F(x) = \ln(x) + c.$$

Let $f : [a, b] \rightarrow \mathbb{R}$ such that F and G are primitives of f on $[a, b]$. Then:

$$F - G = C,$$

because if $H(x) = G(x) - F(x)$, we have:

$$H'(x) = f(x) - f(x) = 0.$$

Since H is defined on $[a, b]$ (a connected interval), it follows that:

$$H(x) = C \implies F(x) = G(x) + C.$$

$$1. \int \frac{dx}{\sqrt{x^2 + 1}} = \operatorname{argsh}(x) + c = F(x).$$

$$2. \int \frac{dx}{\sqrt{x^2 + 1}} = \ln(x + \sqrt{x^2 + 1}) + c_2 = G(x).$$

1.2 Algebraic Rules for Primitives

Let $f, g : [a, b] \rightarrow \mathbb{R}$, and let F be a primitive of f and G a primitive of g . Then:

$$1. \int (f + g)(x) dx = \int f(x) dx + \int g(x) dx.$$

$$2. \int \alpha f(x) dx = \alpha \int f(x) dx \quad (\alpha \in \mathbb{R}).$$

$$3. \int f'(x) dx = f(x) + c \quad (\text{very nice, useful, cute}).$$

- **Proof**

Easy exercise.

If f admits a primitive, it admits infinitely many primitives. For example:

$$\int \frac{dx}{x} = \ln(x) + c,$$

where c is an arbitrary constant. Thus:

$$\int \frac{dx}{x} = \ln(x) + 4,$$

$$\int \frac{dx}{x} = \ln(x) + 5,$$

$$\int \frac{dx}{x} = \ln(x) + e,$$

$$\vdots$$

$$\int \frac{dx}{x} = \ln(x) + c.$$

1.2.1 Definition

The set of all primitives of a function f is called the **indefinite integral** of f . For example:

$$\{\ln(x) + c \mid c \in \mathbb{R}\}.$$

1.3 Techniques for Calculating Primitives

1.3.1 Integration by Parts (IBP)

Let u and v be two functions defined on $[a, b]$ and differentiable. Then:

$$\int u \cdot v' dx = uv - \int u' \cdot v dx.$$

- **Derivation**

From the product rule $(uv)' = u'v + uv'$, we have:

$$\int (uv)' dx = \int u'v dx + \int uv' dx.$$

Thus:

$$uv = \int u'v dx + \int uv' dx,$$

which implies:

$$\int uv' dx = uv - \int u'v dx.$$

1. $\int x^2 \ln(x) dx = \frac{1}{3}x^3 \ln(x) - \frac{1}{3} \int x^3 \cdot \frac{1}{x} dx = \frac{1}{3}x^3 \ln(x) - \frac{1}{9}x^3 + c.$
2. $\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2 \left(x e^x - \int e^x dx \right) = x^2 e^x - 2x e^x + 2e^x + c.$
3. $\int \ln(x) dx$ (let $u' = 1, v = \ln(x)$).

1.3.2 Change of Variables

Let f be a continuous function $f : [a, b] \rightarrow \mathbb{R}$, and let $\phi : [\alpha, \beta] \rightarrow \mathbb{R}$ be differentiable such that $\phi(\alpha) = a$ and $\phi(\beta) = b$. Then:

$$\int f(x) dx = \int f(\phi(t)) \cdot \phi'(t) dt,$$

where $x = \phi(t)$ and $dx = \phi'(t) dt$.

1. $\int \cos(x) \cdot e^{\sin(x)} dx = \int e^{\sin(x)} d(\sin(x)) = e^{\sin(x)} + c.$
Why? $\rightarrow$ Let $t = \sin(x) \rightarrow dt = \cos(x) dx \rightarrow \int e^t dt = e^t + c = e^{\sin(x)} + c.$
2. $\int \frac{e^x}{e^{2x} + 1} dx = \int \frac{e^x}{(e^x)^2 + 1} dx = \arctan(e^x) + c.$

1.3.3 Integrals of the Form $\int \frac{dx}{ax^2+bx+c}$

For integrals of this form, complete the square:

$$ax^2 + bx + c = a \left(\left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2} \right).$$

Given $\frac{4ac-b^2}{4a^2} = \pm M^2$, then:

$$ax^2 + bx + c = a \left(\left(\frac{2ax+b}{2a} \right)^2 \pm M^2 \right) = aM^2 \left(\left(\frac{2ax+b}{2aM} \right)^2 \pm 1 \right).$$

Now, consider the integral:

$$\int \frac{dx}{ax^2 + bx + c} = \frac{1}{aM^2} \int \frac{dx}{\left(\frac{2ax+b}{2aM} \right)^2 \pm 1}.$$

Let $t = \frac{2ax+b}{2aM}$.

$\int \frac{dx}{x^2 + x + 1} = \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$. Here $M = \frac{\sqrt{3}}{2}$. Let $t = x + \frac{1}{2}$, then:

$$\begin{aligned} \int \frac{dt}{t^2 + \frac{3}{4}} &= \int \frac{dt}{\frac{3}{4}\left(\frac{4t^2}{3} + 1\right)} = \frac{4}{3} \int \frac{dt}{\left(\frac{2t}{\sqrt{3}}\right)^2 + 1} \\ &= \frac{2}{\sqrt{3}} \arctan\left(\frac{2t}{\sqrt{3}}\right) + c = \frac{2}{\sqrt{3}} \arctan\left(\frac{2x+1}{\sqrt{3}}\right) + c. \end{aligned}$$

For an integral of the form $I_2(x) = \int \frac{\alpha x + \beta}{ax^2 + bx + c} dx$, we decompose it. One has:

$$\alpha x + \beta = \frac{\alpha}{2a}(2ax + b) + \left(\beta - \frac{\alpha b}{2a}\right) = \frac{\alpha}{2a}(2ax + b) + \frac{2a\beta - \alpha b}{2a}.$$

So the integral becomes:

$$\begin{aligned} I_2(x) &= \int \frac{\frac{\alpha}{2a}(2ax + b) + \frac{2a\beta - \alpha b}{2a}}{ax^2 + bx + c} dx \\ &= \frac{\alpha}{2a} \int \frac{2ax + b}{ax^2 + bx + c} dx + \left(\frac{2a\beta - \alpha b}{2a}\right) \int \frac{dx}{ax^2 + bx + c} \\ &= \frac{\alpha}{2a} \ln|ax^2 + bx + c| + \left(\frac{2a\beta - \alpha b}{2a}\right) I_1(x). \end{aligned}$$

$\int \frac{3x-1}{x^2-x+1} dx$. Note that $(x^2-x+1)' = 2x-1$. We write $3x-1 = \frac{3}{2}(2x-1) + \frac{3}{2}-1 = \frac{3}{2}(2x-1) + \frac{1}{2}$.

$$\begin{aligned} \int \frac{3x-1}{x^2-x+1} dx &= \int \frac{\frac{3}{2}(2x-1) + \frac{1}{2}}{x^2-x+1} dx \\ &= \frac{3}{2} \int \frac{2x-1}{x^2-x+1} dx + \frac{1}{2} \int \frac{dx}{x^2-x+1} \\ &= \frac{3}{2} \ln(x^2-x+1) + \frac{1}{2} \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \frac{3}{2} \ln(x^2-x+1) + \frac{1}{2} \left(\frac{2}{\sqrt{3}} \arctan\left(\frac{x-1/2}{\sqrt{3}/2}\right) \right) + c \\ &= \frac{3}{2} \ln(x^2-x+1) + \frac{1}{\sqrt{3}} \arctan\left(\frac{2x-1}{\sqrt{3}}\right) + c_0. \end{aligned}$$

1.3.4 Integrals of the Form $\int \frac{dx}{\sqrt{ax^2+bx+c}}$

We complete the square in the denominator. Let $\Delta = b^2 - 4ac$.

- If $\Delta > 0$, the quadratic has two distinct roots.

- If $\Delta < 0$, the quadratic has no real roots.

The expression $ax^2 + bx + c$ can be written as $a \left[\left(x + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2} \right]$. The integral becomes $\frac{1}{\sqrt{|a|}} \int \frac{dx}{\sqrt{\left(x + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2}}}$. This leads to inverse hyperbolic or inverse trigonometric functions.

- If $a > 0, \Delta < 0$: $\frac{1}{\sqrt{a}} \operatorname{argsh}(\dots) + c$
- If $a > 0, \Delta > 0$: $\frac{1}{\sqrt{a}} \operatorname{argch}(\dots) + c$
- If $a < 0, \Delta > 0$: $\frac{1}{\sqrt{-a}} \arcsin(\dots) + c$

$\int \frac{dx}{\sqrt{x^2 - x + 1}} = \int \frac{dx}{\sqrt{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}}} = \int \frac{dx}{\sqrt{\left(x - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}}$. Let $t = \frac{x-1/2}{\sqrt{3}/2} = \frac{2x-1}{\sqrt{3}}$. Then $dt = \frac{2}{\sqrt{3}}dx$. The integral becomes:

$$\int \frac{\frac{\sqrt{3}}{2}dt}{\sqrt{\frac{3}{4}t^2 + \frac{3}{4}}} = \int \frac{dt}{\sqrt{t^2 + 1}} = \operatorname{argsh}(t) + c = \operatorname{argsh}\left(\frac{2x-1}{\sqrt{3}}\right) + c.$$

For an integral of the form $I_4(x) = \int \frac{\alpha x + \beta}{\sqrt{ax^2 + bx + c}} dx$, we use a similar decomposition:

$$\alpha x + \beta = \frac{\alpha}{2a}(2ax + b) + \left(\beta - \frac{\alpha b}{2a}\right).$$

The integral becomes:

$$\begin{aligned} I_4(x) &= \frac{\alpha}{2a} \int \frac{2ax + b}{\sqrt{ax^2 + bx + c}} dx + \left(\beta - \frac{\alpha b}{2a}\right) \int \frac{dx}{\sqrt{ax^2 + bx + c}} \\ &= \frac{\alpha}{a} \sqrt{ax^2 + bx + c} + \frac{2a\beta - \alpha b}{2a} \cdot I_3(x). \end{aligned}$$

$\int \frac{5x + 3}{\sqrt{x^2 + 4x + 10}} dx$. We write $5x + 3 = \frac{5}{2}(2x + 4) - 7$.

$$\begin{aligned} \int \frac{5x + 3}{\sqrt{x^2 + 4x + 10}} dx &= \int \frac{\frac{5}{2}(2x + 4) - 7}{\sqrt{x^2 + 4x + 10}} dx \\ &= \frac{5}{2} \int \frac{2x + 4}{\sqrt{x^2 + 4x + 10}} dx - 7 \int \frac{dx}{\sqrt{x^2 + 4x + 10}} \\ &= 5\sqrt{x^2 + 4x + 10} - 7I_3(x). \end{aligned}$$

Now we solve for $I_3(x)$:

$$\begin{aligned} I_3(x) &= \int \frac{dx}{\sqrt{x^2 + 4x + 10}} = \int \frac{dx}{\sqrt{(x+2)^2 + 6}} \\ &= \int \frac{dx}{\sqrt{6}\sqrt{\left(\frac{x+2}{\sqrt{6}}\right)^2 + 1}}. \quad \text{Let } t = \frac{x+2}{\sqrt{6}} \Rightarrow dx = \sqrt{6} dt. \\ &= \int \frac{\sqrt{6} dt}{\sqrt{6}\sqrt{t^2 + 1}} = \int \frac{dt}{\sqrt{t^2 + 1}} = \operatorname{argsh}\left(\frac{x+2}{\sqrt{6}}\right) + C. \end{aligned}$$

So the final answer is $5\sqrt{x^2 + 4x + 10} - 7 \operatorname{argsh}\left(\frac{x+2}{\sqrt{6}}\right) + C$.

1.4 Integration of Rational Functions

To integrate a rational function $\frac{P(x)}{Q(x)}$, we have two cases:

1. $\deg(P) \geq \deg(Q)$: Perform Euclidean division to get $\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$ where $\deg(R) < \deg(Q)$.
2. $\deg(P) < \deg(Q)$: Decompose $\frac{R(x)}{Q(x)}$ into partial fractions.

The integration of $S(x)$ is straightforward. It is sufficient to investigate the second case.

The decomposition depends on the factorization of the denominator $Q(x)$.

- Simple roots: $Q(x) = (x - \alpha_1) \dots (x - \alpha_n) \implies \frac{P(x)}{Q(x)} = \sum \frac{A_i}{x - \alpha_i}$.
- Multiple roots: $Q(x) = (x - \alpha_1)^{m_1} \dots (x - \alpha_r)^{m_r} \implies \frac{P(x)}{Q(x)} = \sum_{i=1}^r \sum_{j=1}^{m_i} \frac{A_{ij}}{(x - \alpha_i)^j}$.

$f(x) = \frac{x^2+1}{x^2-1}$. By division, $f(x) = 1 + \frac{2}{x^2-1}$. We decompose the fraction:

$$\frac{2}{x^2 - 1} = \frac{A}{x - 1} + \frac{B}{x + 1} = \frac{A(x + 1) + B(x - 1)}{(x - 1)(x + 1)} = \frac{(A + B)x + (A - B)}{x^2 - 1}.$$

Solving $A + B = 0$ and $A - B = 2$ gives $A = 1, B = -1$. Thus:

$$\begin{aligned} f(x) &= 1 + \frac{1}{x - 1} - \frac{1}{x + 1}. \\ \int f(x) dx &= x + \ln|x - 1| - \ln|x + 1| + C. \end{aligned}$$

$g(x) = \frac{2x+1}{x^2-1}$.

$$g(x) = \frac{2x + 1}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1} = \frac{(A + B)x + (A - B)}{x^2 - 1}.$$

Solving $A + B = 2$ and $A - B = 1$ gives $A = 3/2$, $B = 1/2$. Thus:

$$g(x) = \frac{3/2}{x-1} + \frac{1/2}{x+1}.$$

$$\int g(x)dx = \frac{3}{2} \ln|x-1| + \frac{1}{2} \ln|x+1| + C.$$

1.4.1 Decomposition of Rational Functions

A rational function is a ratio of two polynomials $f(x) = \frac{P(x)}{Q(x)}$, defined for all x where $Q(x) \neq 0$.

Case 1: $\deg(P) \geq \deg(Q)$

Perform polynomial division: $P(x) = Q(x)S(x) + R(x)$, where $\deg(R) < \deg(Q)$. Then:

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}.$$

Case 2: $\deg(P) < \deg(Q)$

Use partial fraction decomposition based on the factors of $Q(x)$.

1. Distinct Linear Factors: $Q(x) = (x - a_1) \dots (x - a_n)$

$$\frac{P(x)}{Q(x)} = \frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \dots + \frac{A_n}{x - a_n}.$$

2. Repeated Linear Factors: $Q(x) = (x - a_1)^{m_1} \dots (x - a_n)^{m_n}$

$$\frac{P(x)}{Q(x)} = \sum_{i=1}^n \left(\frac{A_{i1}}{x - a_i} + \frac{A_{i2}}{(x - a_i)^2} + \dots + \frac{A_{im_i}}{(x - a_i)^{m_i}} \right).$$

3. Irreducible Quadratic Factors: $Q(x) = (x^2 + b_1x + c_1) \dots (x^2 + b_nx + c_n)$

$$\frac{P(x)}{Q(x)} = \sum_{i=1}^n \frac{A_i x + B_i}{x^2 + b_i x + c_i}.$$

4. Repeated Irreducible Quadratic Factors: $Q(x) = (x^2 + b_1x + c_1)^{m_1} \dots (x^2 + b_nx + c_n)^{m_n}$

$$\frac{P(x)}{Q(x)} = \sum_{i=1}^n \sum_{j=1}^{m_i} \frac{A_{ij}x + B_{ij}}{(x^2 + b_i x + c_i)^j}.$$

1.4.2 Decomposition Steps

1. Factor the denominator completely.
2. Write the general form of partial fractions.
3. Multiply through by the LCD (Least Common Denominator).
4. Solve for unknown coefficients.
5. Substitute back into partial fractions and integrate.

- For linear factors: $\int \frac{A}{x-a} dx = A \ln |x-a| + C$.
- For repeated linear factors ($n \neq 1$): $\int \frac{A}{(x-a)^n} dx = \frac{A}{(1-n)(x-a)^{n-1}} + C$.
- For quadratic factors: $\int \frac{Ax+B}{x^2+bx+c} dx$ is solved by splitting the numerator, leading to a logarithm and an arctangent term.

Notes

- Always check if the numerator degree is greater than or equal to the denominator degree first.
- For irreducible quadratics, complete the square if necessary.
- The method of undetermined coefficients is used to find the constants.
- Integration of partial fractions typically results in logarithmic and arctangent terms.

1.4.3 Repeated Irreducible Quadratic Factors

To integrate $\int \frac{Ax+B}{(ax^2+bx+c)^k} dx$, we split the integral:

$$\begin{aligned} \int \frac{Ax+B}{(ax^2+bx+c)^k} dx &= \frac{A}{2a} \int \frac{2ax+b}{(ax^2+bx+c)^k} dx + \left(B - \frac{Ab}{2a}\right) \int \frac{dx}{(ax^2+bx+c)^k} \\ &= \frac{A}{2a(1-k)} (ax^2+bx+c)^{1-k} + \left(B - \frac{Ab}{2a}\right) J_k, \end{aligned}$$

where $J_k = \int \frac{dx}{(ax^2+bx+c)^k}$. This integral can be solved using a recurrence relation derived from integration by parts, after completing the square and making a substitution. For the part $\left(B - \frac{Ab}{2a}\right) J_k$, we refer to the method in Section 1.3.3.

1.5 Further Integration Techniques

1.5.1 Integrals Involving Roots and Rational Substitutions

For integrals of the form $\int R(x, x^{k/l}, x^{m/n}, \dots, x^{r/s}) dx$.

Let $\alpha = \text{lcm}(l, n, \dots, s)$, and set $x = t^\alpha$. Then $dx = \alpha t^{\alpha-1} dt$. This transforms the integrand into a rational function of t .

$\int \frac{dx}{\sqrt{x}(1 + \sqrt[4]{x})}$. Here, $\alpha = \text{lcm}(2, 4) = 4$. Substitute $x = t^4$, $dx = 4t^3 dt$:

$$\int \frac{4t^3 dt}{t^2(1+t)} = 4 \int \frac{t}{1+t} dt = 4 \int \left(1 - \frac{1}{1+t}\right) dt = 4(t - \ln|1+t|) + C.$$

Substitute back $t = \sqrt[4]{x}$.

$\int \frac{\sqrt[3]{1+x}}{\sqrt[4]{1+x} + 1} dx$. Let $1+x = t^{12}$, so $\alpha = \text{lcm}(3, 4) = 12$. Then $dx = 12t^{11} dt$.

$$\int \frac{t^4}{t^3 + 1} \cdot 12t^{11} dt = 12 \int \frac{t^{15}}{1 + t^3} dt.$$

Using polynomial division: $t^{15} = (t^{12} - t^9 + t^6 - t^3 + 1)(t^3 + 1) - 1$. The integral is $12 \int \left(t^{12} - t^9 + t^6 - t^3 + 1 - \frac{1}{1+t^3}\right) dt$.

1.5.2 Integrals Involving $\sqrt{ax^2 + bx + c}$ (Euler Substitutions)

For integrals of the form $\int R(x, \sqrt{ax^2 + bx + c}) dx$.

1. **If $a > 0$:** Substitute $\sqrt{ax^2 + bx + c} = \pm\sqrt{a}x + t$. Squaring and solving for x yields a rational function of t .
2. **If $c > 0$:** Substitute $\sqrt{ax^2 + bx + c} = xt \pm \sqrt{c}$. Squaring and solving for x yields a rational function of t .
3. **If $a < 0$ and $\Delta = b^2 - 4ac > 0$:** Let α, β be roots of $ax^2 + bx + c = 0$. Substitute $\sqrt{ax^2 + bx + c} = \sqrt{a(x - \alpha)(x - \beta)} = (x - \alpha)t$.

1.5.3 Integrals of Rational Trigonometric Functions

For integrals of the form $\int R(\cos x, \sin x) dx$.

Let $t = \tan(x/2)$. Then:

$$x = 2 \arctan t, \quad dx = \frac{2 dt}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\int \frac{1+\cos x}{\sin x} dx = \int \frac{1+\frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}} \cdot \frac{2 dt}{1+t^2} = \int \frac{\frac{2}{1+t^2}}{\frac{2t}{1+t^2}} \cdot \frac{2 dt}{1+t^2} = \int \frac{1}{t} dt = \ln |t| + C = \ln \left| \tan \frac{x}{2} \right| + C.$$

- If $R(-\sin x, \cos x) = -R(\sin x, \cos x)$ (odd in sine), substitute $t = \cos x$.
- If $R(\sin x, -\cos x) = -R(\sin x, \cos x)$ (odd in cosine), substitute $t = \sin x$.
- If $R(-\sin x, -\cos x) = R(\sin x, \cos x)$ (even in both), substitute $t = \tan x$.

1.6 Advanced Trigonometric Integration

1.6.1 Trigonometric Identities

1. **Angle Sum Identities:** $\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$, $\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$.
2. **Pythagorean Identity:** $\cos^2(a) + \sin^2(a) = 1$.
3. **Power-Reducing (Double Angle) Identities:** $\cos^2(a) = \frac{1+\cos(2a)}{2}$, $\sin^2(a) = \frac{1-\cos(2a)}{2}$.
4. **Product-to-Sum Identities:**

$$\cos(a) \cos(b) = \frac{\cos(a-b) + \cos(a+b)}{2}$$

$$\sin(a) \sin(b) = \frac{\cos(a-b) - \cos(a+b)}{2}$$

$$\sin(a) \cos(b) = \frac{\sin(a+b) + \sin(a-b)}{2}$$

1.6.2 Integration Examples

$$\int \cos^{2024}(x) \sin(x) dx. \text{ Let } u = \cos(x), du = -\sin(x)dx.$$

$$= \int -u^{2024} du = -\frac{u^{2025}}{2025} + C = -\frac{\cos^{2025}(x)}{2025} + C.$$

$$\begin{aligned} \int \cos^4(x) dx &= \int \left(\frac{1 + \cos(2x)}{2} \right)^2 dx = \frac{1}{4} \int (1 + 2\cos(2x) + \cos^2(2x)) dx \\ &= \frac{1}{4} \int \left(1 + 2\cos(2x) + \frac{1 + \cos(4x)}{2} \right) dx \\ &= \frac{1}{4} \int \left(\frac{3}{2} + 2\cos(2x) + \frac{1}{2}\cos(4x) \right) dx \\ &= \frac{1}{4} \left(\frac{3}{2}x + \sin(2x) + \frac{1}{8}\sin(4x) \right) + C = \frac{3}{8}x + \frac{1}{4}\sin(2x) + \frac{1}{32}\sin(4x) + C. \end{aligned}$$

$$\begin{aligned} \int \sin(17x) \cos(21x) dx &= \frac{1}{2} \int (\sin(17x + 21x) + \sin(17x - 21x)) dx \\ &= \frac{1}{2} \int (\sin(38x) - \sin(4x)) dx \\ &= \frac{1}{2} \left(-\frac{1}{38} \cos(38x) + \frac{1}{4} \cos(4x) \right) + C \\ &= -\frac{1}{76} \cos(38x) + \frac{1}{8} \cos(4x) + C. \end{aligned}$$

For $\int \frac{\cos(x) + \sin(x)}{7 + \cos(x)} dx$. An alternative approach:

$$\begin{aligned} \int \frac{\cos(x) + \sin(x)}{7 + \cos(x)} dx &= \int \frac{\cos(x) + 7 - 7 + \sin(x)}{7 + \cos(x)} dx \\ &= \int \left(1 - \frac{7}{7 + \cos(x)} + \frac{\sin(x)}{7 + \cos(x)} \right) dx \\ &= x - \int \frac{7}{7 + \cos(x)} dx - \int \frac{-\sin(x)}{7 + \cos(x)} dx \\ &= x - \ln |7 + \cos(x)| - 7 \int \frac{dx}{7 + \cos(x)}. \end{aligned}$$

The last integral is solved with $t = \tan(x/2)$:

$$\int \frac{dx}{7 + \cos(x)} = \int \frac{\frac{2dt}{1+t^2}}{7 + \frac{1-t^2}{1+t^2}} = \int \frac{2dt}{7(1+t^2) + (1-t^2)} = \int \frac{2dt}{6t^2 + 8} = \int \frac{dt}{3t^2 + 4}.$$

This evaluates to an arctangent function.

A. Additional Resources

Here are some helpful resources for practicing and understanding integration.

- **100 Integrals Video (blackpenredpen):** A fast-paced marathon of solving 100 integrals, showcasing a huge variety of techniques. Great for review and inspiration.
https://youtu.be/dgm4-3-Iv3s?si=lo3Qa2tkmr_KZazC
- **Online Integral Calculator:** A powerful tool for checking your work. It provides step-by-step solutions for most integrals.
<https://www.integral-calculator.com/>
- **Prof. Benbachir Playlist:** A comprehensive playlist covering indefinite integrals (primitives) in detail. (In French/Arabic)
<https://youtube.com/playlist?list=PLIznM09WVcDRWD2uK52cM0nBK4SfLnCNy>